STM32 Test platform introduction:

This set of STM32 test programs use the development board of the ALIENTEK, as follows:

Development board: MiniSTM32, Elite STM32, Explorer STM32F4, Apollo STM32F4/F7

MCU: STM32F103RCT6, STM32F103ZET6, STM32F407ZGT6, STM32F429IGT6

(Corresponding to the above development boards)

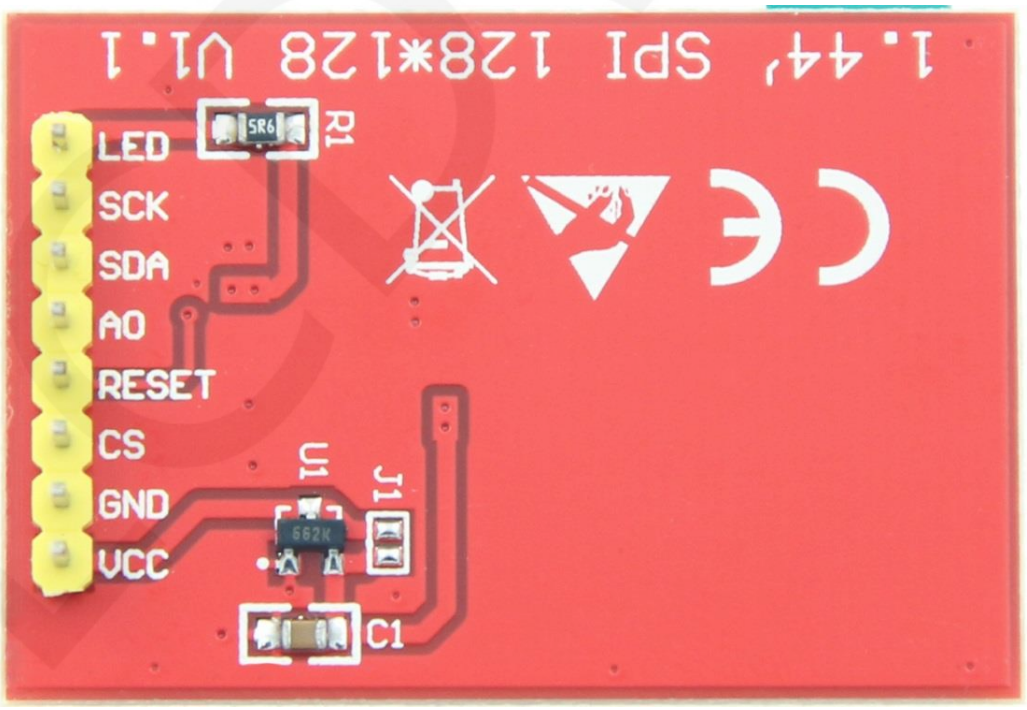
Main frequency: 72M, 72M, 168M, 180M9 (Corresponding to the above MCU)

Crystal frequency: 8M, 8M, 8M, 25M (Corresponding to the above MCU)

These four types of development boards have similar functions, but there are differences in the main frequency and performance, and have no effect on running the test program.

Wiring instructions:

Because the pin positions of different development boards are different, and the external pins are reserved differently (some development boards do not have externally required pins). In order to facilitate wiring, the wiring pins of each development board are inconsistent. described as follows:



Pin silkscreen picture

STM32F103RCT6 microcontroller test program wiring instructions			
Number	**Module Pin**	**Corresponding to MiniSTM32 development board wiring pin**	**Remarks**
1	**VCC**	5V/3.3V	LCD power supply is positive (3.3V~5V)
2	**GND**	GND	LCD Power ground
3	**CS**	PB11	LCD selection control signal
4	**RESET**	PB12	LCD reset control signal
5	**A0**	PB10	LCD register / data selection control signal
6	**SDA**	PB15	LCD SPI bus write data signal
7	**SCK**	PB13	LCD SPI bus clock signal
8	**LED**	PB9	LCD backlight control signal (high level lighting, if you do not need control, please connect 3.3V)

STM32F103ZET6 microcontroller test program wiring instructions			
Number	**Module Pin**	**Corresponding to Elite STM32 development board wiring pin**	**Remarks**
1	**VCC**	5V/3.3V	LCD power supply is positive (3.3V~5V)
2	**GND**	GND	LCD Power ground
3	**CS**	PB11	LCD selection control signal
4	**RESET**	PB12	LCD reset control signal
5	**A0**	PB10	LCD register / data selection control signal
6	**SDA**	PB15	LCD SPI bus write data signal
7	**SCK**	PB13	LCD SPI bus clock signal

8	**LED**	PB9	LCD backlight control signal (high level lighting, if you do not need control, please connect 3.3V)

STM32F407ZGT6 microcontroller test program wiring instructions			
Number	**Module Pin**	**Corresponding to Explorer STM32F4 development board wiring pin**	**Remarks**
1	**VCC**	5V/3.3V	LCD power supply is positive (3.3V~5V)
2	**GND**	GND	LCD Power ground
3	**CS**	PB15	LCD selection control signal
4	**RESET**	PB12	LCD reset control signal
5	**A0**	PB14	LCD register / data selection control signal
6	**SDA**	PB5	LCD SPI bus write data signal
7	**SCK**	PB3	LCD SPI bus clock signal
8	**LED**	PB13	LCD backlight control signal (high level lighting, if you do not need control, please connect 3.3V)

STM32F429IGT6 microcontroller test program wiring instructions			
Number	**Module Pin**	**Corresponding to Apollo STM32F4/F7 development board wiring pin**	**Remarks**
1	**VCC**	5V/3.3V	LCD power supply is positive (3.3V~5V)
2	**GND**	GND	LCD Power ground
3	**CS**	PD11	LCD selection control signal

4	**RESET**	PD12	LCD reset control signal
5	**A0**	PD5	LCD register / data selection control signal
6	**SDA**	PF9	LCD SPI bus write data signal
7	**SCK**	PF7	LCD SPI bus clock signal
8	**LED**	PD6	LCD backlight control signal (high level lighting, if you do not need control, please connect 3.3V)

Demo function description:

1. This test program contains four test procedures for STM32 MCU, namely: STM32F103RCT6, STM32F103ZET6, STM32F407ZGT6, STM32F429IGT6;
2. Each MCU test program includes two functional tests: software SPI and hardware SPI;
3. When using the software SPI function or hardware SPI function of each MCU, the wiring pin definition is the same, but the initialization is different;
4. Please follow the above wiring instructions to find the corresponding development board and MCU for wiring;
5. This set of tests supports display switching in four directions. For details, see the display direction switching instructions.
6. After touching the display direction, the touch screen needs to be calibrated;
7. This set of test procedures contains the following test items:
   A. 3D menu display test;
   B. the main interface display test
   C. simple brush test;
   D. rectangular drawing and filling test;
   E. circular drawing and filling test;
   F. triangle drawing and filling test;
   G. English display test;

H. Chinese display test;

I. picture display test;

J. rotating display test;

8. this module does not have a touch function, please remove the touch screen handwriting test items;

Display direction switching instructions:

Find the macro definition USE_HORIZONTAL in lcd.h as shown below:

```
//////////////////////////////////////用户配置区//////////////////////////////////////
#define USE_HORIZONTAL     0//定义液晶屏顺时针旋转方向  0-0度旋转，1-90度旋转，2-180度旋转，3-270度旋转
```

USE_HORIZONTAL 0 //0° Rotate

USE_HORIZONTAL 1 //90° Rotate

USE_HORIZONTAL 2 //180° Rotate

USE_HORIZONTAL 3 //270° Rotate